Week 1: Revision of statistical parameter estimation concepts

F79MA Statistical Models A

Heriot-Watt University

Notation and definitions

A *random sample* (of size n) from a distribution is a set of n independent random variables:

$$\underline{X} = (X_1, X_2, \ldots, X_n) \ ,$$

each coming from that distribution and having probability density function (or probability mass function) $f(x;\theta)$, where θ is a parameter (or *vector of parameters*). Note that we use **upper case letters** to denote random variables.

We use **lower case letters** to denote the actual observations, or realised values, in an experiment:

$$\underline{x} = (x_1, x_2, \ldots, x_n) \ .$$

A *statistic* is a random variable $T = T(\underline{X})$ which is a function of the data $\underline{X}$, but which does not depend on θ.

An *estimator* is a statistic $T = T(\underline{X})$ such that the value of the realisation $t = T(\underline{x})$ is to be used as an estimate of θ.

A note on frequentist and Bayesian perspectives

In the second half of the semester we will see that there are multiple philosphical approaches to interpreting notions such as expectations. Let us briefly give an idea of this here and relate it to the definitions above.

Suppose X is a random variable. Then we know that if X is discrete then $\mathbf{E}(X)$ is given by

$$\mathbf{E}(X) = \sum_x xf(x),$$

where the sum is taken over all possible values of x and $f(x) = \mathbf{P}(X = x)$. Similarly, if X is continuous we have

$$\mathbf{E}(X) = \int_{x=-\infty}^{\infty} xf(x),$$

where $f(x)$ is the probability density function. But what does $\mathbf{E}(X)$ mean?

A **frequentist** views probability theory as being related to describing the possible outcomes of many identical experiments. For example, a frequentist would see $\mathbf{E}(X)$ as being the average value of the outcomes of a sufficiently large number of identical and independent experiments, or equivalently the average value of a sufficiently large random sample. In other words, a frequentist believes that

$$\mathbf{E}(X) \approx \frac{1}{n}\sum_{j=1}^{n} x_j$$

for $(x_1, \ldots, x_n)$ a large random sample.

A **Bayesian** by contrast would view $\mathbf{E}(X)$ in some sense as the value we would expect a **single** future realisation to take (even if that value is not itself possible such as non-integer expectation for an integer-valued random variable) based on their prior beliefs and any evidence available. We will make this more precise in Week 7 onwards. For the remainder of the first half of the course we will generally assume a frequentist position as this is the classical approach to statistics.

Returning to the notion of a statistic $T = T(\underline{X})$, a frequentist would interpret $\mathbf{E}(T)$ as being the average value of T over a sufficiently large number m of independent samples of size n. In other words, if we have large number m of independent samples of size n, where the jth sample could be written as $\underline{x}_j = (x_{j1}, \ldots, x_{jn})$, a frequentist would believe

$$\mathbf{E}(X) \approx \frac{1}{m}\sum_{j=1}^{m} T(\underline{x}_j).$$

If, for example, $T(\underline{x}_j$ is the sample mean of $\underline{x}_j$, then this means

$$\mathbf{E}(X) \approx \frac{1}{m}\sum_{j=1}^{m}\left(\frac{1}{n}\sum_{i=1}^{n} x_{ji}\right) = \frac{1}{mn}\sum_{j=1}^{m}\sum_{i=1}^{m} x_{ji}.$$

Example 1.1

Let $\underline{X} = (X_1, X_2, \ldots, X_n)$ be a random sample from a Poisson distribution with parameter (mean) μ. Consider the following four estimates (estimators) for μ:

(1) $T_1 = \overline{X} = \frac{1}{n}\sum X_i$ (the sample mean);

(2) $T_2 = X_1$ (the first observation);

(3) $T_3 = \sum w_i X_i$ where w_i's are pre-specified real numbers;

(4) $T_4 = \frac{1}{n}\sum X_i^2$ (mean of squares of X_i's).

To explore these estimates, consider the first and the second order moments:

(1)

$$\begin{aligned}\mathbf{E}(T_1) &= \mathbf{E}(\overline{X}) = \mu \quad \text{(good?)},\\ \mathbf{Var}(T_1) &= \mathbf{Var}(\overline{X}) = \sigma^2/n = \mu/n.\end{aligned}$$

Note that the variance gets smaller as n increases and tends to 0 as $n \to \infty$ (good?)

(2)

$$\begin{aligned}\mathbf{E}(T_2) &= \mathbf{E}(X_1) = \mu,\\ \mathbf{Var}(T_2) &= \mathbf{Var}(X_1) = \mu\end{aligned}$$

(3)

$$\begin{aligned}\mathbf{E}(T_3) &= \sum w_i \mathbf{E}(X_i) = \mu \sum w_i \qquad (= \mu \text{ only if } \textstyle\sum w_i = 1),\\ \mathbf{Var}(T_3) &= \sum w_i^2 \mathbf{Var}(X_i) = \mu \sum w_i^2.\end{aligned}$$

(4)

$$\begin{aligned}\mathbf{E}(T_4) &= \frac{1}{n}\sum \mathbf{E}(X_i^2) = \frac{1}{n}\cdot n(\mu+\mu^2) = \mu+\mu^2,\\ \mathbf{Var}(T_4) &= \frac{1}{n^2}\sum \mathbf{Var}(X_i^2).\end{aligned}$$

Properties of estimators

The following concepts are central to how we characterise the accuracy of an estimator:

Bias

The bias of an estimator $T = T(\underline{X})$ is defined by

$$B(T) = \mathbf{E}\,(T-\theta)\,.$$

T is *unbiased* for θ if $\mathbf{E}(T-\theta) = 0$ (or $\mathbf{E}(T) = \theta$).

In other words, roughly speaking from a frequentist perspective this means the average value of T over many experiments should equal θ, where the size of the sample in each experiment is not important.

T is *asymptotically unbiased* for θ if

$$B(T) \to 0 \qquad \text{as } n \to \infty\,.$$

In other words, roughly speaking from a frequentist perspective this means the average value of T over many experiments each with large sample sizes should equal θ.

Unbiasedness seems to be desirable but it is not essential and, by itself, does not guarantee a good estimator. Asymptotic unbiasedness is essential for a good estimator.

Mean square error

The mean square error is defined to be

$$MSE(T) = \mathbf{E}\left[(T-\theta)^2\right] = \mathbf{Var}(T) + [B(T)]^2 .$$

For estimators T_1 and T_2 we say that T_1 is *more efficient* than T_2 if

$$MSE(T_1) < MSE(T_2)$$

Note that a biased estimator might be more efficient than an unbiased estimator.

Consistency

T is (mean-square) consistent for θ if

$$MSE(T) \to 0 \qquad \text{as } n \to \infty .$$

T is consistent in probability if

$$\forall \epsilon > 0, \qquad \mathbf{P}\left(|T-\theta| > \epsilon\right) \to 0 \qquad \text{as } n \to \infty .$$

For a 'good' estimator, we usually require *at least* that $\mathbf{E}(T) \to \theta$ and $\mathbf{Var}(T) \to 0$ as $n \to \infty$.

Ideally we should like to find estimators that are optimally efficient in that they have the smallest possible variance whilst being asymptotically unbiased – this will be discussed in Week 3.

Example 1.1 revisited

We have already considered the first and the second order moments of the estimators, we now focus on *unbiasedness*, *consistency*, and *efficiency*.

(1) $\mathbf{E}(T_1) = \mathbf{E}(\overline{X}) = \mu$, therefore T_1 is unbiased.

$\mathbf{Var}(T_1) = \mu/n \to 0$ as $n \to \infty$, and T_1 is consistent.

(2) T_2 is also unbiased, but not consistent: $\mathbf{Var}(T_2) = \mu = \text{constant} > 0$, so the variance does not tend to 0.

Furthermore, $\mathbf{Var}(T_1) < \mathbf{Var}(T_2)$ (except when $n = 1$). Therefore, T_1 is more efficient than T_2.

(3) T_3 is unbiased only if $\sum w_i = 1$, i.e., if it is a weighted average.

*E*xercise: Assume that $\sum w_i = 1$ so that T_3 is unbiased. Show that $\mathbf{Var}(T_3)$ is minimised by setting $w_i = 1/n$ for all $i = 1, \ldots, n$.

(**Hint**: Set $w_n = 1 - \sum_{i=1}^{n-1} w_i$, and take partial derivatives with respect to $i = 1, \ldots, n-1$ to find the minimum of $\mathbf{Var}(T_3)$.)

(4) T_4 is not unbiased, i.e., it is biased (unless $\mu = 0$).

Clearly, $\mathbf{Var}(T_4) \to 0$ as $n \to \infty$. This is fine, but $MSE(T_4)$ does not tend to zero due to the bias: $B(T_4) = \mathbf{E}(T_4) - \mu = \mu^2 = \text{constant} > 0$, for any n.

Constructing estimators: the method of moments

The method of moments is perhaps the simplest approach to finding estimators.

General approach: Given a random sample $(X_1, \ldots, X_n)$, find parameter(s) $\widetilde{\theta}$ which make the (theoretical) population moments match the (observed) sample moments.

For models with a single parameter, we often match the theoretical population mean with the sample mean. We find $\widetilde{\theta}$ that solves

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i = \mu(\theta)\,.$$

The value of $\widetilde{\theta}$ that solves this is the *method-of-moments estimator* (MME).

It is possible to derive estimators from other moments too. One can even consider estimators based on negative, fractional, and logarithmic moments. This is useful, for example, for random variables that do not have a mean, a variance, or higher order moments (e.g., the Cauchy random variable).

Example 1.2

Let $\underline{X} = (X_1, \ldots, X_n)$ be a random sample from exponential distribution with parameter λ, Exp(λ). This means that its pdf is $f(x;\lambda) = \lambda e^{\lambda x}$ for $x \geq 0$ and $f(x;\lambda) = 0$ for $x < 0$.

MME: The population mean is $\mathbf{E}(X) = 1/\lambda$ and the sample mean is $\overline{X}$. So the MME comes from $\overline{X} \approx 1/\lambda$, and so $\widetilde{\lambda} = 1/\overline{X}$.

Some data: Assume that $n = 10$ and $\underline{x} = (0.2, 0.4, 0.5, 0.8, 1.2, 1.5, 1.6, 1.6, 3.1, 5.3)$. Then $\overline{X} = 1.62$ and $\widetilde{\lambda} = 1/1.62 = 0.617$.

How good is the MME estimate? Since

$$\mathbf{E}(\widetilde{\lambda}) = \mathbf{E}\left(\frac{1}{\overline{X}}\right) \neq \frac{1}{\mathbf{E}(\overline{X})},$$

the MME is biased.

We can find an expression for the bias. For that, we need the distribution of $\overline{X}$. We use the following key steps:

- We use the fact that the Exponential random variable is a special case of the Gamma random variable, i.e., $X_i \sim \text{Exp}(\lambda) = \text{Gamma}(1, \lambda)$.
- We use the properties of sums of Gamma random variables to obtain that $Y = \sum X_i \sim \text{Gamma}(n, \lambda)$.
- From the linearity of the expectation operator

  $$\mathbf{E}\left(\frac{1}{\overline{X}}\right) = \mathbf{E}\left(\frac{n}{Y}\right) = n\int_0^\infty \frac{f(y)}{y}dy\,.$$

  where $f(y)$ is the pdf of Y, given by $f(y) = \frac{\lambda^n}{\Gamma(n)} y^{n-1}\exp\{-\lambda y\}$.

- Because $\int_0^\infty f(y)dy = 1$, integrals of the form $\int_0^\infty y^{n-1}\exp\{-\lambda y\}dy = \frac{\Gamma(n)}{\lambda^n}$. Therefore (check this at home),

$$\mathbf{E}\left(\frac{1}{\overline{X}}\right) = \mathbf{E}\left(\frac{n}{Y}\right) = n\int_0^\infty \frac{f(y)}{y}dy = \frac{n}{n-1}\lambda.$$

  A similar approach may be used for finding **Var** and **MSE**.

Moreover, if we have two parameters, θ_1 and θ_2, then we will have to solve two simultaneous equations, usually

$$\begin{aligned} \overline{X} &= \mu(\theta_1, \theta_2)\,, \\ S^2 &= \sigma^2(\theta_1, \theta_2)\,. \end{aligned}$$

Exercise 1.3

Let $\underline{X}$ be a random sample from Gamma(k, λ), with pdf

$$f(x; k, \lambda) = \frac{\lambda^k}{\Gamma(k)}x^{k-1}e^{-\lambda x}, \quad \text{for } x \geq 0.$$

Show that $\mathbf{E}(X) = k/\lambda$ and $\mathbf{Var}(X) = k/\lambda^2$ and hence that the MMEs are $\widetilde{\lambda} = \frac{\overline{X}}{S^2}, \widetilde{k} = \frac{(\overline{X})^2}{S^2}$. The properties of these estimate are not easy to investigate analytically (but we can explore them by using numerical simulation methods, more on this later).

The method of moments is simple, (usually) sensible and very general. All it requires are expressions for population moments ($\mathbf{E}(X)$, $\mathbf{Var}(X)$ etc) in terms of parameters. However, MMEs can be inefficient and sometimes 'nonsensical', as we will see at the exercise class.

Interval estimation

A point estimator T on its own is of limited value. We need some idea of its *accuracy*. Its moments, $\mathbf{E}(T)$ and $\mathbf{Var}(T)$, give some idea of how useful it will be as an estimator of θ, but a better way to communicate to non-statisticians is by using *interval estimation*.

Interval estimation involves defining an interval $I(\underline{X}) = (\theta_L(\underline{X}), \theta_U(\underline{X}))$ such that

$$\mathbf{P}\left(\theta \in I(\underline{X})\right) = 1 - \alpha\,,$$

where α is suitably small (often 0.05). Such an interval $I(\underline{X})$ is called a $100(1-\alpha)\%$ confidence interval for θ.

Note that $\underline{X}$ is a random vector, so $I(\underline{X})$ is a *random interval*. When we obtain a particular sample $\underline{x}$ we **must not write**

$$\mathbf{P}(\theta \in I(\underline{x})) = 1 - \alpha\,,$$

because there are no random quantities here!

Example 1.4

Let $\underline{X}$ be a random sample from a normal distribution $\mathrm{N}(\mu, 1)$. We will develop a 95% CI for the unknown mean μ.

We know that $\overline{X} \sim N(\mu, 1/n)$, or

$$\sqrt{n}(\overline{X} - \mu) \sim N(0, 1)\,.$$

Hence, we can write

$$\mathbf{P}(-1.96 < \sqrt{n}(\overline{X} - \mu) < 1.96) = 0.95\,.$$

This means

$$\mathbf{P}\left(\mu - \frac{1.96}{\sqrt{n}} < \overline{X} < \mu + \frac{1.96}{\sqrt{n}}\right) = 0.95\,.$$

This is a probability statement about the random variable $\overline{X}$.

However, we may rearrange the inequality to obtain

$$\mathbf{P}\left(\overline{X} - \frac{1.96}{\sqrt{n}} < \mu < \overline{X} + \frac{1.96}{\sqrt{n}}\right) = 0.95\,,$$

and so

$$I(\underline{X}) = \left(\overline{X} - \frac{1.96}{\sqrt{n}}, \overline{X} + \frac{1.96}{\sqrt{n}}\right)$$

is a 95% confidence interval for μ.

*N*ote: the CI is symmetric w.r.t. $\overline{X}$, we can write $\overline{X} \pm \frac{1.96}{\sqrt{n}}$ as 95% confidence limits for μ.

Some data: Assume $n = 5$ and $\underline{X} = (2.8, 1.5, 3.0, 2.3, 3.2)$. Then $\overline{X} = 2.56$ (also, for the future, $S = 0.68$), and our 95% CI for μ is $2.56 \pm 1.96/\sqrt{5}$, i.e. $(1.68, 3.44)$.

Terminology: $\mathbf{sd}(\overline{X}) = \sqrt{\mathbf{Var}(\overline{X})}$ is called the *standard error* of the estimator $\overline{X}$.

In Example 1.4 above, the confidence interval from the data is $(1.68, 3.44)$. *This is a random sample from a population of intervals,* 95% *of which contain the 'true' value of* μ. So, we are 95% confident that it contains μ.

Confidence intervals capture the essence of *classical* (or frequentist) statistics; namely that the process generating the data is repeatable and that probabilities are the frequencies with which events occur over long sequences of repetitions.

Confidence intervals are not unique

In Example 1.4, we could also have used

$$\mathbf{P}\left(-1.88 < \sqrt{n}\left(\overline{X} - \mu\right) < 2.05\right) = 0.95$$

when defining a confidence interval.

This is not symmetric, but still leads to a valid CI (which will be wider than the one we obtained in Example 1.4). For confidence intervals for the normal mean it is natural

to exploit symmetry and use the 'equal-tailed' CI, as this is the narrowest interval that can be obtained.

Sometimes we wish to report, for example, the smallest plausible value of a quantity – in other words, a lower bound. Here it may be appropriate to report a CI of the form (θ_L, ∞). Similarly, one could also construct a CI to reflect an upper bound.

Pivotal quantities

Pivotal quantities give a *general method* for finding confidence intervals.

A *pivotal quantity* or *pivot* is a function of our data $\underline{X}$ and the parameter θ such that

- its distribution is known, and
- its distribution does not depend on θ.

In Example 1.4 we used the pivotal quantity

$$\sqrt{n}\left(\overline{X} - \mu\right) \sim \mathrm{N}(0,1)\,.$$

In general we write a probability statement

$$\mathbf{P}\left(z_1 < Z(\underline{X}, \theta) < z_2\right) = 1 - \alpha\,,$$

for a pivotal quantity $Z(\underline{X}, \theta)$. Then, as long as $Z(\underline{X}, \theta)$ is a monotonic function of θ, we can rearrange the inequalities to give a $100(1-\alpha)\%$ confidence interval for θ by writing

$$\mathbf{P}\left(\theta_L(\underline{X}) < \theta < \theta_U(\underline{X})\right) = 1 - \alpha\,.$$

Example 1.5

Let $\underline{X}$ be a random sample of size n from a normal distribution $\mathrm{N}(\mu, \sigma^2)$, where both μ and σ^2 are unknown. We find: (i) a 95% CI for μ, and (ii) a 95% CI for σ^2.

(i) What sampling distribution do we know? The first candidate is

$$\frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} \sim \mathrm{N}(0,1),$$

but we cannot use this as a pivot as in Example 1.4, due to the presence of the unknown parameter σ.

However, we also have

$$\frac{\sqrt{n}(\overline{X} - \mu)}{S} \sim t_{n-1}\,,$$

where $S = \sqrt{S^2}$ is the sample standard deviation. We can use this as a pivot:

$$\mathbf{P}\left(-t_{n-1}(2.5\%) < \frac{\sqrt{n}(\overline{X} - \mu)}{S} < t_{n-1}(2.5\%)\right) = 0.95\,,$$

where $t_{n-1}(2.5\%)$ comes from tables.

Then a 95% CI for μ (using lower case letters to denote observed quantities) is

$$\overline{x} \pm \frac{t_{n-1}(2.5\%)s}{\sqrt{n}}\,.$$

Note that this is symmetric, as in Example 1.4 above.

(ii) We also know that

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

We can use this as a pivot (noting that it does not involve μ!):

$$\mathbf{P}\left(\chi^2_{n-1}(97.5\%) < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{n-1}(2.5\%)\right) = 0.95$$

Then we invert the inequalities and get

$$\mathbf{P}\left(\frac{(n-1)S^2}{\chi^2_{n-1}(2.5\%)} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{n-1}(97.5\%)}\right) = 0.95.$$

Therefore, a 95% confidence interval for σ^2 is

$$\left(\frac{(n-1)s^2}{\chi^2_{n-1}(2.5\%)}, \frac{(n-1)s^2}{\chi^2_{n-1}(97.5\%)}\right).$$

Note that

- the confidence interval is *not* symmetric, so we cannot write confidence limits in the form "something ± something";
- it uses "equal tails" but it may be not be the shortest CI;
- for a CI for σ, just take square roots.